

VITA MUTATUR, NON TOLLITUR

"Life is changed, not taken away." - 1st passage of the Roman Catholic rite of Mass for the Dead.

When life slips away -

when the flesh

and bones

and feathers

and skin

and leaves

decay,

we say they are gone.

Some will protest, saying that it was unjust.

Others will declare satisfaction as an enemy is pulled behind the veil.

Still others will somberly greet the occasion, knowing that the life gone was one filled with love.

In all circumstances, life returns to where it was borrowed.

The ash and dust,

the molecular building blocks

must be relinquished and the reservoir replenished.

Do not fear.

The custodians of the reservoir are gentle.

Thin and delicate appendages deconstruct the form.

Quiet.

Gracious.

They work hard, these clavigers.

To disentangle life from the matrix it was temporarily molded to is no easy feat.

From the rudiments which remain after the task is complete, life sprouts forth.

The question remains.

Is life really gone, or does it simply reconstitute given the right conditions?

ONLY THE FUNGI KNOW.

Penicillium chrysogenum

Scientific names given to organisms are often complex and unintuitive to the layperson, however the scientific name of this particular species of mold is instantly recognizable. This fungal species produces beta-lactam compounds, which act as powerful antibacterials on gram positive bacteria (Gadd & Sarislan, 2019). Since the serendipitous discovery of penicillin by Sir Alexander Fleming in 1928 (Abraham et al., 1941), this fungus has contributed immensely to human health, reducing once life-threatening bacterial infections such as rheumatic fever and pneumonia (Abraham et al., 1941) to trivial matters resolved with a dose or two of penicillin. Further research on this already fruitful test subject revealed the mold's capacity to produce other compounds useful to humans, used in genomics and viroscics, as agricultural agents, and more (Fierro et al., 2022).

Cordyceps militaris

The life cycle of this genus is truly unique. It infects an insect and takes over the nervous system, compelling the insect to act in the fungus's best interests (Tjebkjo et al., 2021). It is endemic to, and culturally significant to, China, India, Tibet and Nepal, where consumption was historically used to resolve cardiovascular, respiratory, liver and kidney diseases (Holiday, 2017). While consumption of the *Cordyceps* fungus is novel to the field of western medicine, further research is showing that the bioactive compounds within the fruiting bodies of *Cordyceps* sp. (specifically *C. militaris*) have unique properties which could fuel advances in countless health fields (Holiday, 2017). In particular, the molecule cordycepin shows promise as an antiviral for SARS-CoV-2 (Holiday, 2017).

Fomitopsis betulina

Fungi have accompanied humans along our evolutionary journey from hunters and gatherers to the modern *Homo sapiens*, providing small comforts to early hominids. There is strong evidence for this relationship between fungi and early humans found with Europe's oldest human mummy, Otzi the iceman (Peintner et al., 1998). In the preserved man's leather bag, two *F. betulina* fruiting bodies strung on leather strips (Peintner et al., 1998). The exact use of these will probably never be determined (some current theories suggest spiritual uses, food or firestarter), however it is clear that the fungi were important enough to the neolithic human to be clutched even in death.

Cryomyces antarcticus

Hidden within the rocks of Antarctica's McMurdo Dry Valleys, the microscopic fungus *C. antarcticus* has become an unlikely model organism for space exploration research (Pacelli et al., 2017). It is remarkably resistant to ultraviolet radiation, desiccation, and extreme temperature fluctuations, conditions similar to those found in outer space and Mars (Oufri et al., 2019). As a result, *C. antarcticus* has been included in several astrobiology experiments conducted on the International Space Station (Oufri et al., 2019). Colonies of the fungus were exposed to vacuum, cosmic radiation, and simulated Martian environments for over a year, with many cells remaining intact with their DNA largely preserved (Pacelli et al., 2017). These research findings help scientists evaluate the limits of life in space and on Mars.

Panaeolus africanus

While it will never be determined when exactly humans started including hallucinogenic substances in rituals, it is certain that altered states of consciousness were an important part of human evolution, shaping religion and culture. One such fungus thought to contribute to religious thought is *P. africanus* (Winkelman, 2019). Often found growing in elephant dung, this fungus contains inconsistent amounts of the hallucinogenic compounds psilocybin and psilocin (Strauss et al., 2023). *P. africanus* has shaped both our past and future, as psilocybin and psilocin have been successfully used to treat mental health conditions such as major depressive disorder and PTSD (Goodwin et al., 2023).

Moniliophthora perniciosa

This species of fungus strikes fear into the likes of Willy Wonka, the Easter Bunny, and children all over the world. Known for causing "Witch's Broom disease", this fungus attacks and decimates cocoa trees, which produce the raw components of chocolate (Farquharson, 2014). Endemic to South America, *M. perniciosa* is blamed for the collapse of the cacao industry in Brazil in the late 1980s, which resulted in the death of majority of the country's cacao trees and countless livelihoods (Farquharson, 2014). This also affected the global chocolate landscape, driving up production in other nations such as Côte d'Ivoire (Farquharson, 2014). Fungi such as *M. perniciosa* have the potential to dictate global economic trends and transform lives (for better or for worse), reinforcing the need for further research.

Laetisporium basilioides

Wildfires leave landscapes blackened and seemingly devoid of life, but some organisms are uniquely adapted to thrive in their aftermath. One such species is *L. basilioides*, a fungus from Australia that often grows after bushfires. This species survives extreme heat by forming a sclerotium, a hardened mass of fungal tissue that remains protected beneath the soil during fires (2018). When conditions stabilize after the burn, the fungus produces fruiting bodies and are one of the first organisms visible in recovering ecosystems (McMullan-Fisher et al., 2014). By breaking down charred plant material, *L. basilioides* helps recycle nutrients back into the soil. In this way, fungi play an important role in restoring soil quality after wildfires.

References

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POSTER BY MAYA (KYRIN) SOOK YAN LOH. 10/03/2026